Genomic offset (and GEA) for the study of biological invasions

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Outline

Local adaptation and Genomic offset

Biological invasions

Simulation study : can we use genomic offset to inform invasion risk ?

Empirical study : what GEA and genomic offset can tell us about the past and future invasions of *D. suzukii* ?

General conclusions and perspectives

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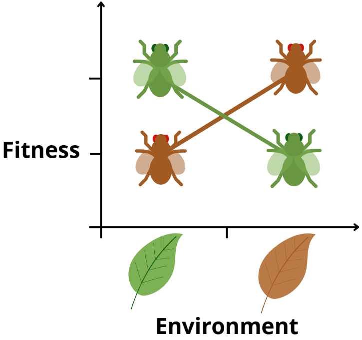
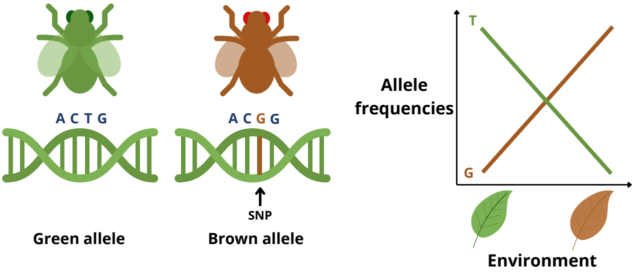
Local adaptation

Populations develop traits that give them a higher average fitness in their local environment than populations from other environments



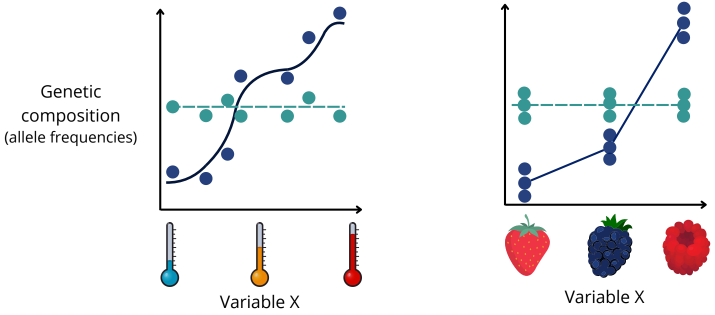
Local adaptation

Follows (among others) from the spatial correspondence between adaptive genetic variation and environmental variation



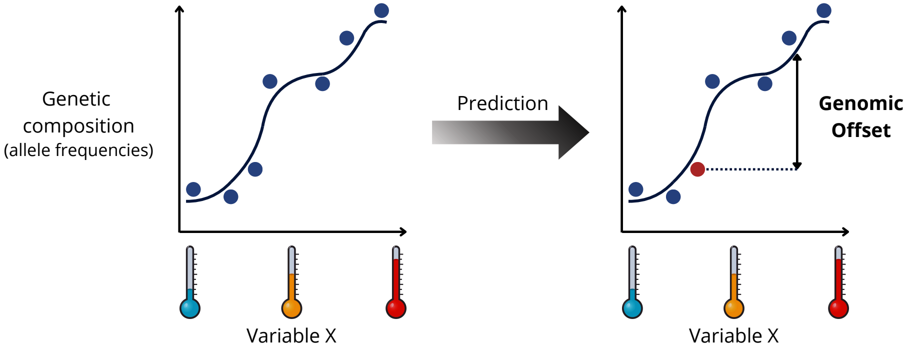
Detect local adaptation

GEA = Genome Environment Association



- From genome-wide genetic data, identify markers/environmental variables with a role in local adaptation

The Genomic Offset (Fitzpatrick et Keller, 2015)



Genomic Offset (GO) = a measure of the risk of (mal)adaptation

- Temporal (future environmental conditions)
- **Spatial (other geographic areas)**

Important assumptions

- Training populations locally adapted
- Adaptation through pre-existing variants
- GEA correctly captures the link between allele frequencies and the environnement.
- Invariant genome $\leftrightarrow$ environment relationship along time.

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Context

Invasive species

- Increasingly frequent in recent years as a result of globalization (Seebens, 2021a, Hulme, 2021)
- Potentially harmful
- Example of rapid and successful environmental change

→ How to explain and predict their success in “new” environments ?

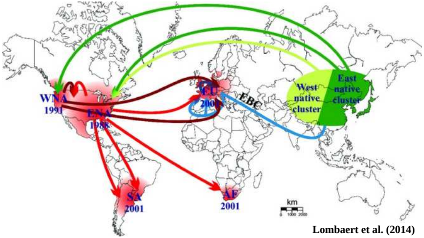
The invasion paradox

- Reduced number of founding individuals
  - Ex : asian hornet (Arca et al., 2015)
  - Reduction of genetic diversity, inbreeding, ... → reduced adaptive potential

- Adaptation to an "unknown" environment : pre-adaptation ?

Potential insights from genomics

- **Neutral aspects** : reconstruction of invasion history



Potential insights from genomics

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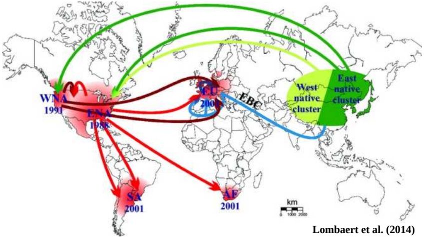


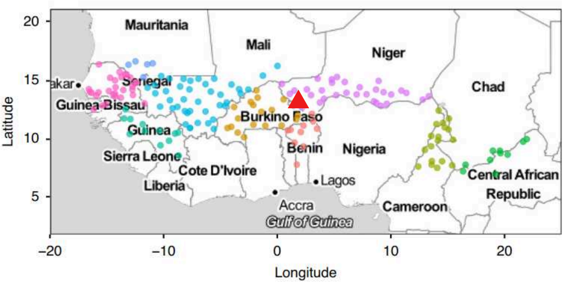
- **Adaptive aspects** :
    - Genome scans for selection : identification of (potential) genomic regions related to invasion success.
    - Genomic Offset : evaluate (mal)adaptation to new geographic areas → **predict invasion risk**.

Genomic offset assumptions & biological invasions

- Training populations locally adapted ↔ Rather use the native area
- Adaptation through pre-existing variants ↔ OK for short term prediction
- GEA correctly captures the link between allele frequencies and environnement.
- Invariant genome ↔ environment relationship along time. OK for short term prediction

Empirical validation of the genomic offset

Link between genomic offset and fitness establihed so far **only for instantaneous spatial changes** (common gardens, e.g. Rhoné *et al.* 2020, Fitzpatrick *et al.* 2021)



↔ relevant for **biological invasions**.

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Invasion process

Transport

Invasion process

Transport → **Introduction**

Invasion process

Transport → **Introduction** → **Establishment**

Survival and reproduction sufficient to establish stable and self-sustaining population(s)

Invasion process

Transport → **Introduction** → **Establishment** → **Invasion**

Invasive species whose individuals disperse, survive and reproduce on multiple sites

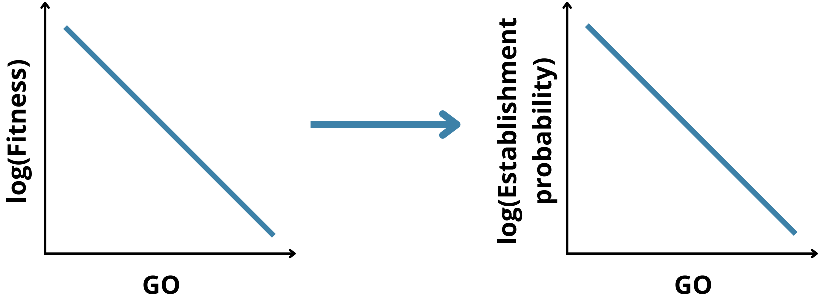
Invasion process

Transport → **Introduction** → **Establishment** → **Invasion**

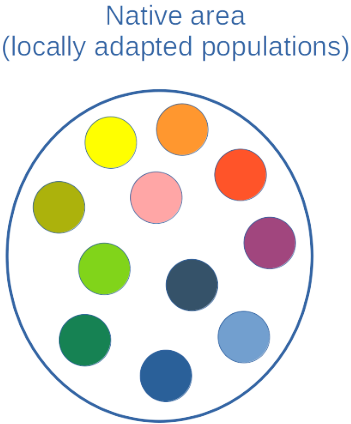
The last stage before a species is considered as invasive

Sufficient fitness to maintain one/several stable population(s)

Hypothesis :



Simulation design



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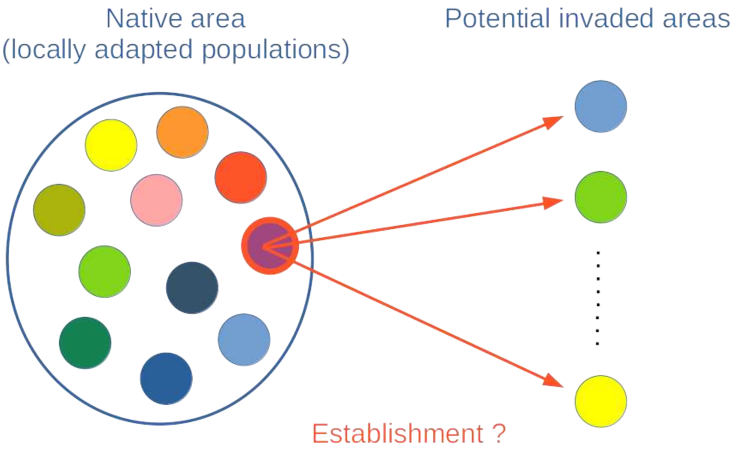

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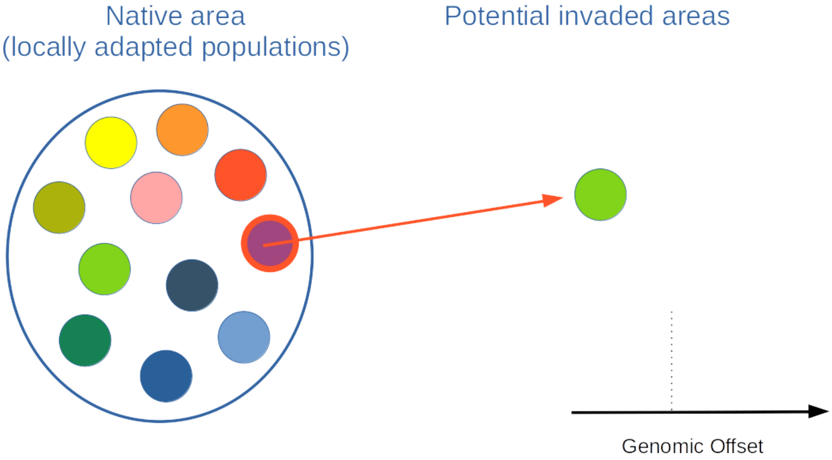

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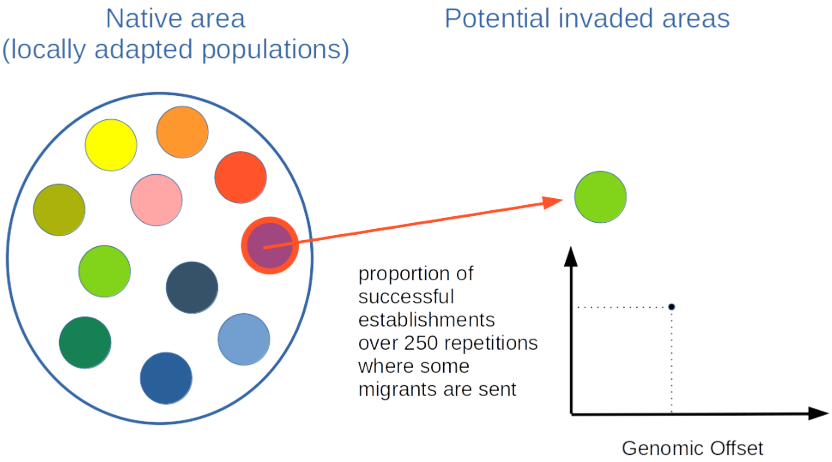

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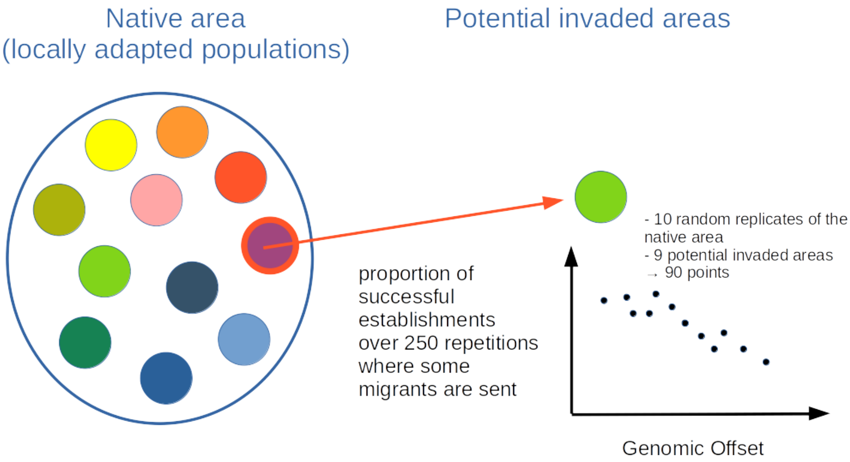

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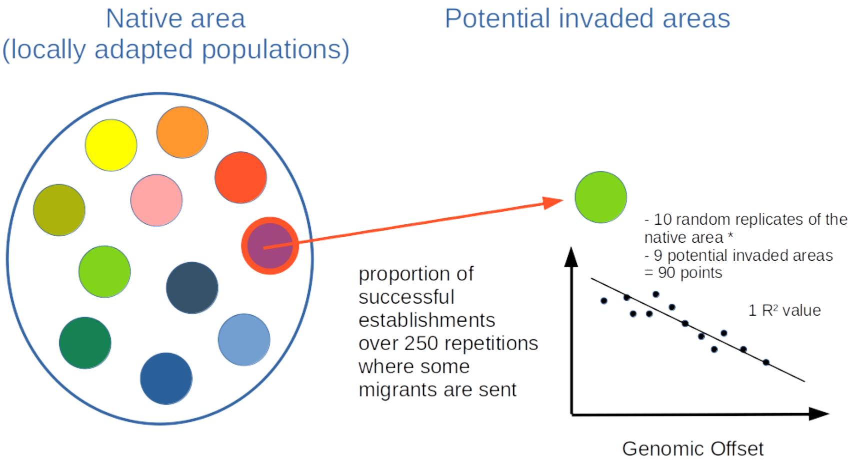

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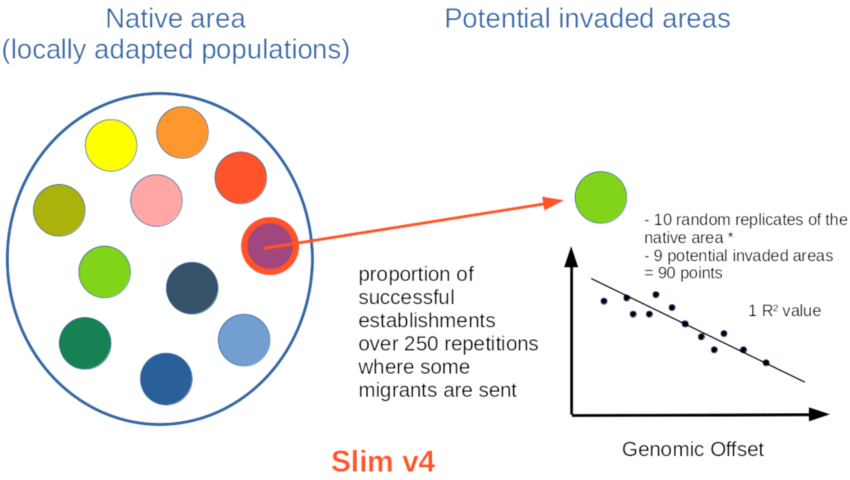
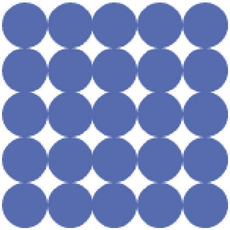


Simulation details : native area

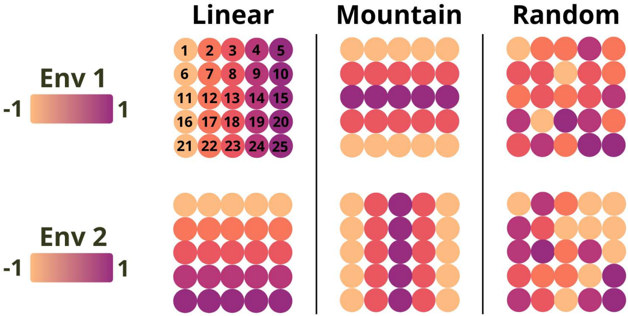
Inspired from Làruson et al (2022)

- Population grid (5x5), stepping stone Wright-Fisher model
- 1000 individuals/population
- High (0.05) or low (0.005) migration rate
- 3000 generations



Simulation details : native area

- 2 environmental variables
- Adaptation *via* two QTLs (Quantitative Trait Loci)
    - Σ effect sizes = phenotype
    - fitness = phenotype $\leftrightarrow$ environment
- 3 different environment types



Simulations details : invaded area

- Panmictic population characterized by the 2 env. variables
  ↪ **9 possible environments to invade** : (-1,0,1) × (-1,0,1)
- 10 ou 100 invading individuals
  ↪ **3 possible source populations** : -1/-1, 0/0, 1/1
- Non constrained population size ("Non-WF") :
  - Overlapping generations
  - Death ≠ Reproduction
- Simulation stops when the population :
  - goes extinct
  - **goes established**
    ↪ reaches 50 000 individuals
    ↪ survives for 100 generations

Genomic offset methods

Geometric GO (Gain et al. 2023)

- Linear (regression)

env. var. regression coefficients

$$\mathbf{Y} = \mathbf{X}\mathbf{B}' + \mathbf{W} + \mathbf{E}$$

allele freq. latent factors

- Correction for neutral structure :
  → Latent factors (*LFMM*)
  → Allele frequencies covariance (*Baypass*)

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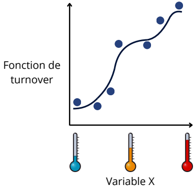
Gradient Forest (Fitzpatrick & Keller, 2015)

- Non Linear (Random Forest)



- Allele frequencies corrected by LFMM
  - Optimized

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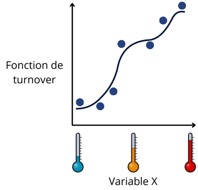


- Allele frequencies corrected by LFMM
  - Optimized

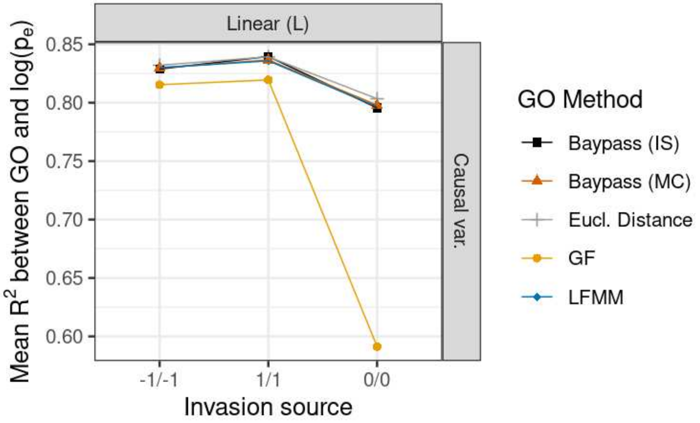
+ Euclidian environnemental distance (no GEA)

Genomic offset computation

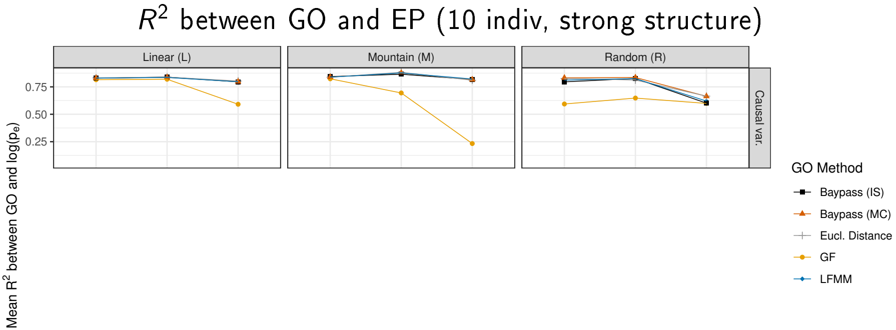
- **Variables** :
  - 2 causal
  - 2 causal + 8 counfounding
  - PCs of the 2+8
- **SNPs** :
  - **All** (global MAF $\geq 1\%$)
  - Top 10% higly differentiated (XtX^* statistic)

Results in an "ideal" case

R^2 between GO and EP (10 indiv, strong structure)



Results with only causal variables



Performance reduced by confounding variables

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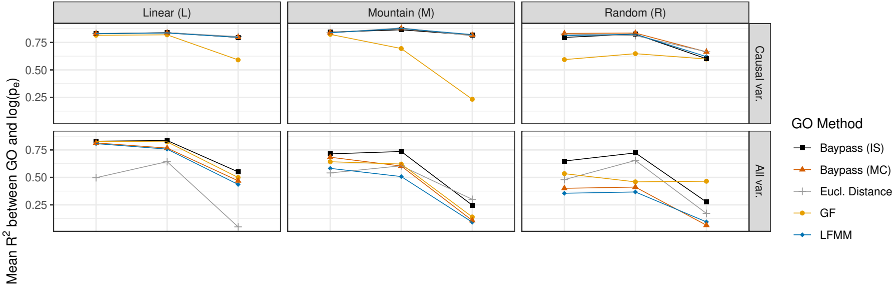
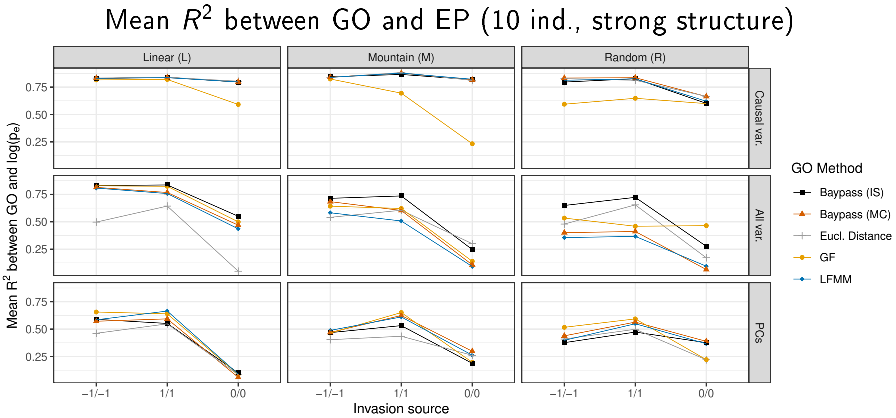


PCs mitigates the difference between methods



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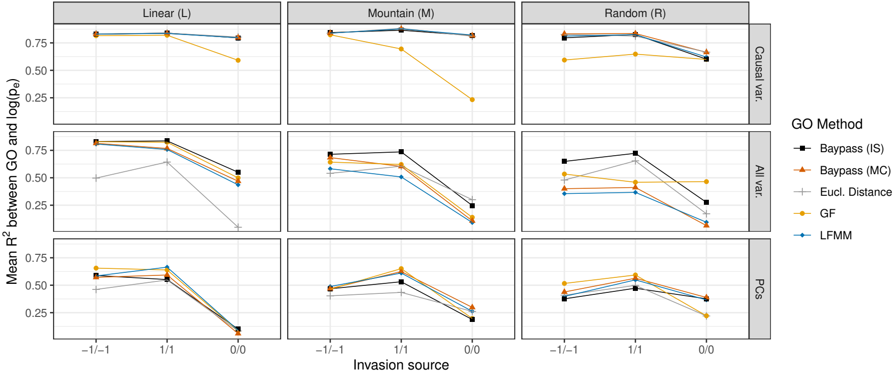
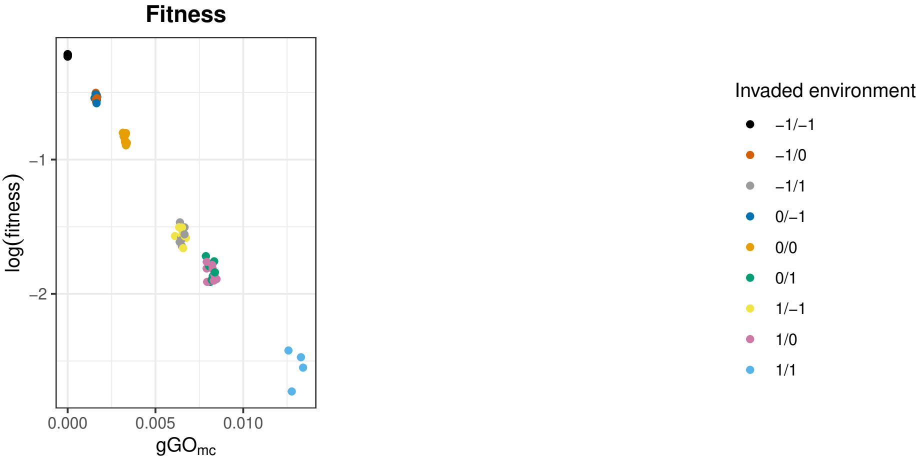
Mean R^2 between GO and EP (10 ind., strong structure)



Similar results with weak structure

Effect of propagule size

100 individuals → strong relationship with fitness



Effect of propagule size

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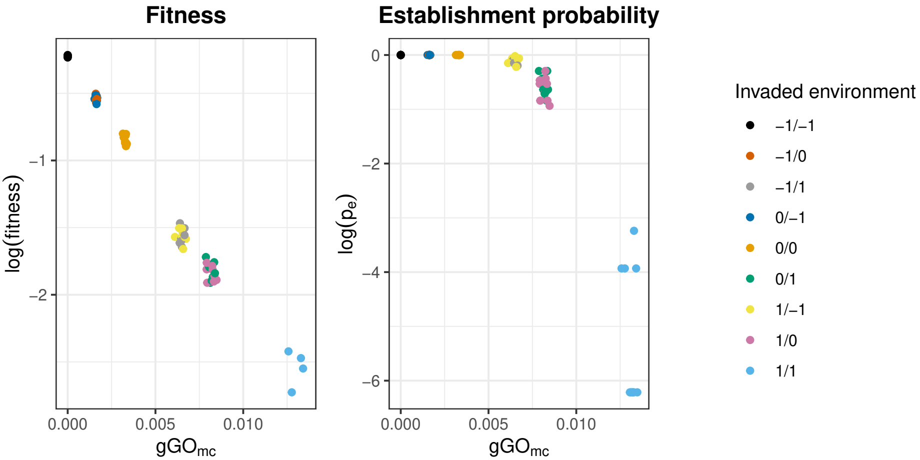
But adaptive challenge buffered



Conclusions

Relationship between GO and establishment probability ?

- Strong correlation between GO and establishment probability
- Robust to adaptation strenght (=structuration) in the native area
- Disturbed by confounding variables and environmental complexity
- Weakened / destroyed when many migrants

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Methods performances ?

- Use of univariate methods or PCs recommended
- Geometric GO performs better

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Some questions remains ...

- Influence of admixture ? Interpretability of absolute GO values ?

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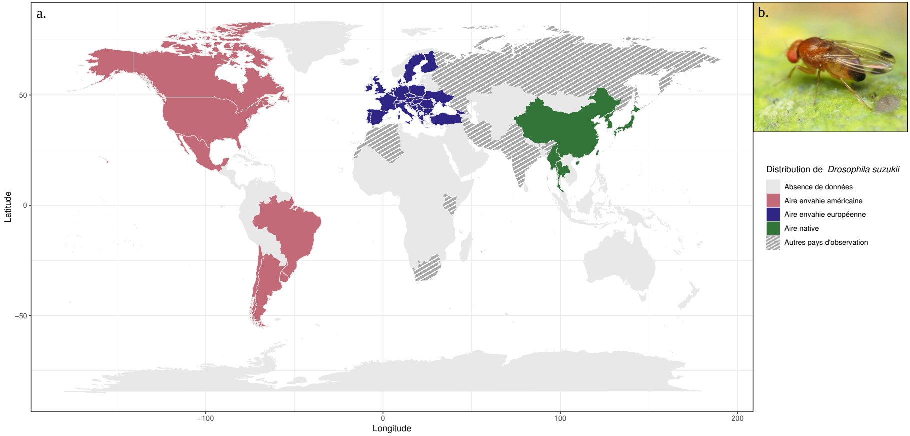
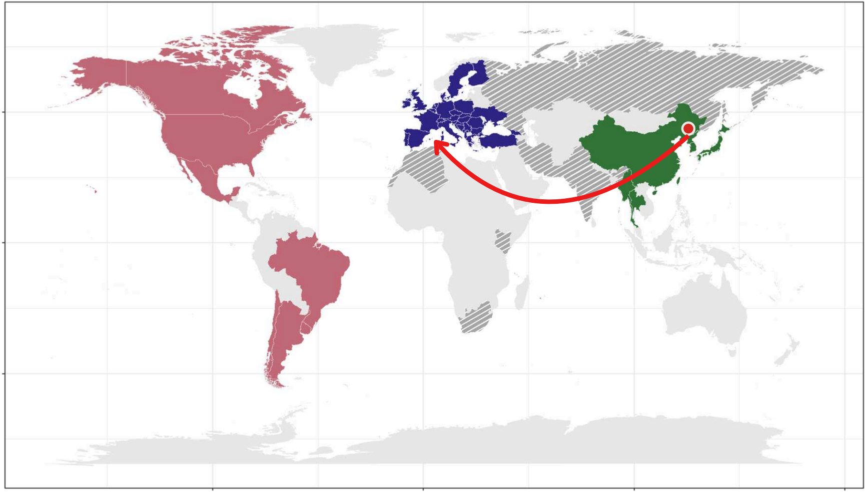
D. suzukii : an invasive species...



...with 2 independant invasions... (Adrion et al.,2014, Fraimout et al., 2017)



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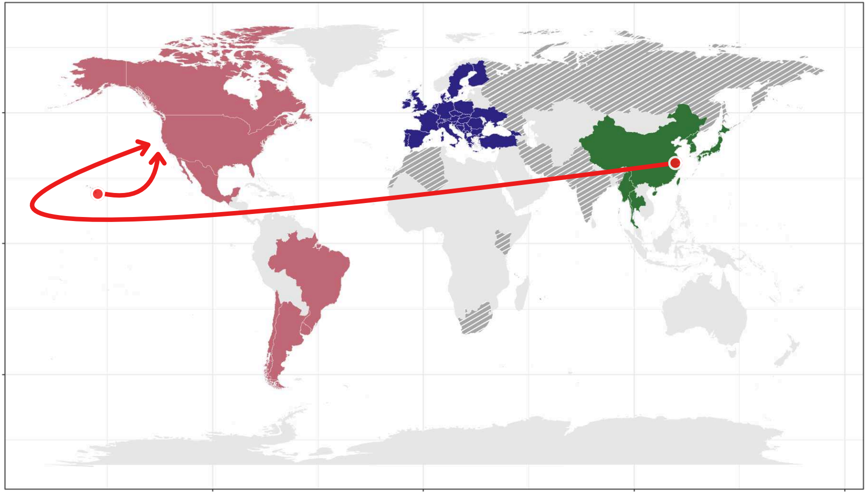


… and a crop pest

- Sclerified ovipositor : can lay eggs in ripe or ripening fruits
- A wide range of host plants (120 species reported in Europe)

Motivations

Research aims

- To understand the *adaptive dimension* of *D. suzukii*'s worlwide invasion : adaptive challenge ? key genes ?
- To try **predicting** regions at risk of invasion

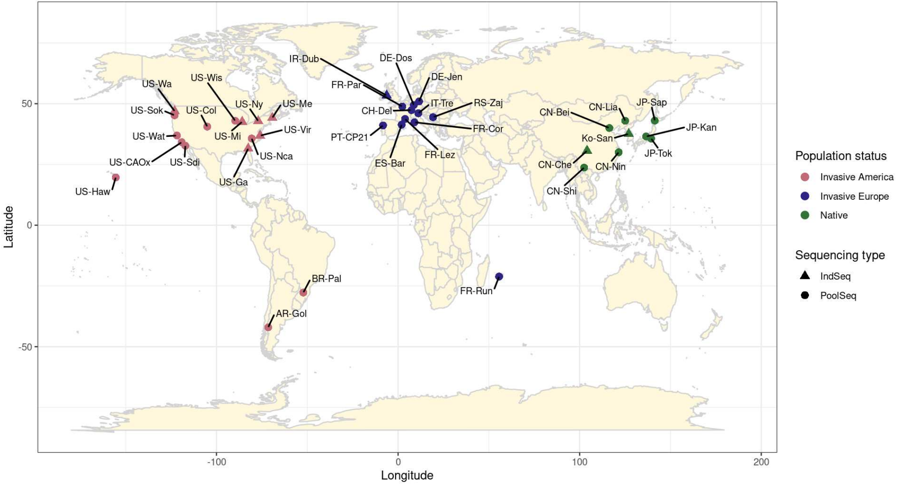
A rich dataset

- 21 + 2 + 1 Pool-Seq samples (Olazcuaga et al. 2020, Feng et al. 2024, Sario et al. 2024)
- 8 Ind-Seq samples (Lewald et al. 2021)
- 4 Pool-Seq samples + 1 Ind-Seq sample (CBGP)

→ Analysis of combined data with Baypass

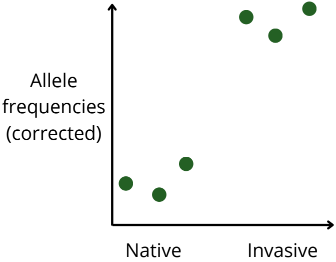
New data

37 populations ≈ 4 200 000 SNPs identified with a new assembly



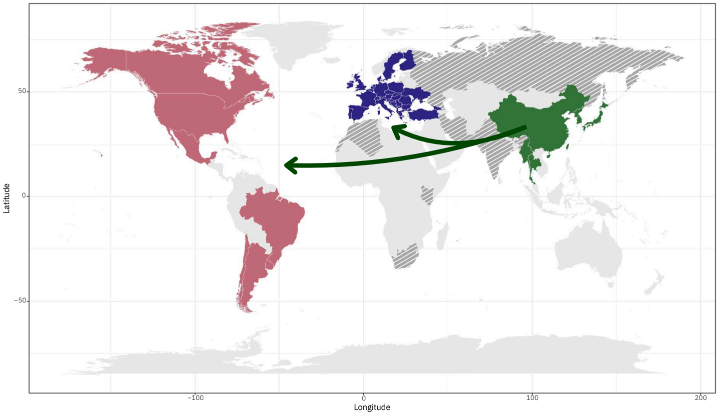
Genomic regions associated to invasive status ?

1. Candidate SNPs from Baypass C_2 index (Olazcuaga et al., 2020).



2. Candidate genomic regions (i.e. with an excess of candidate SNPs) from the local-score approach (Fariello et al., 2017).
3. Genes in these regions (NCBI) and associated biological processes (orthologous genes in *Drosophila melanogaster*)

Genomic regions associated to invasive status ?



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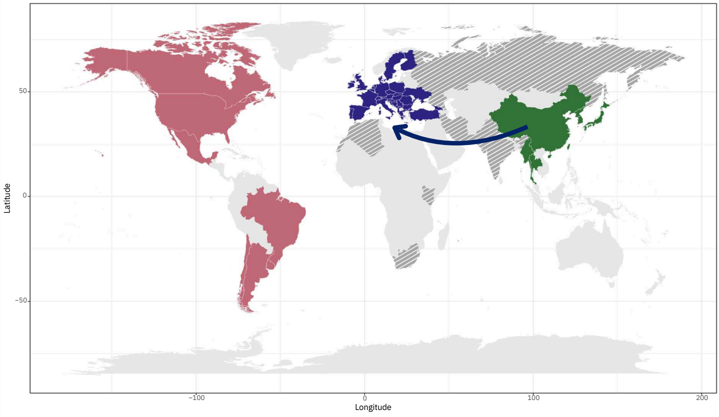

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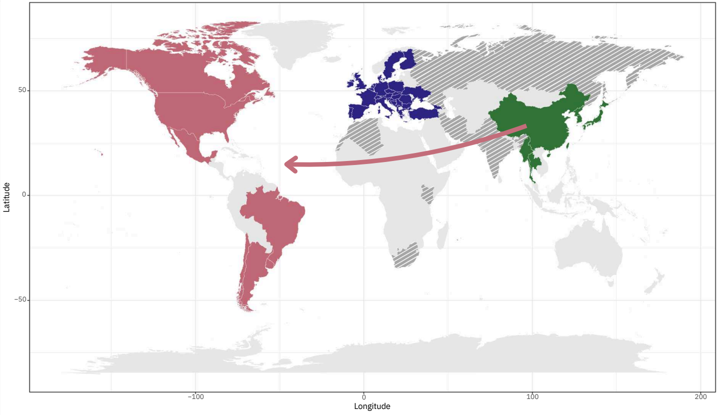

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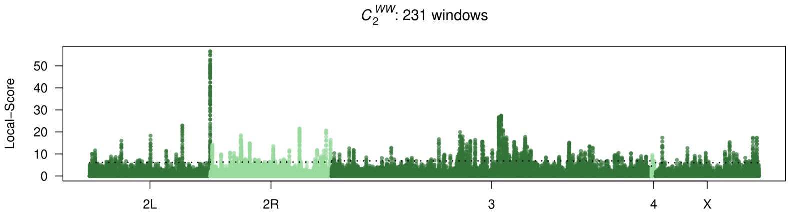
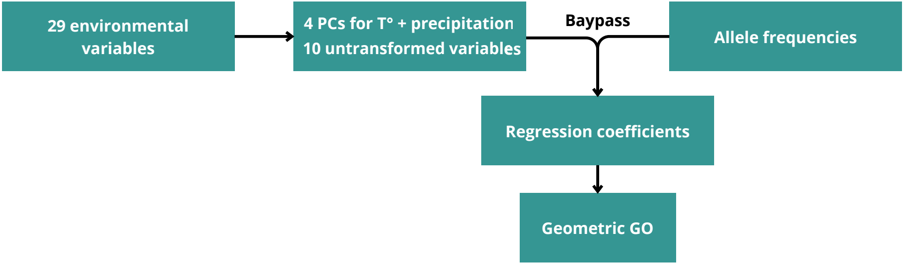


396 candidate genes (in 457 regions, all contrasts) related to e.g.

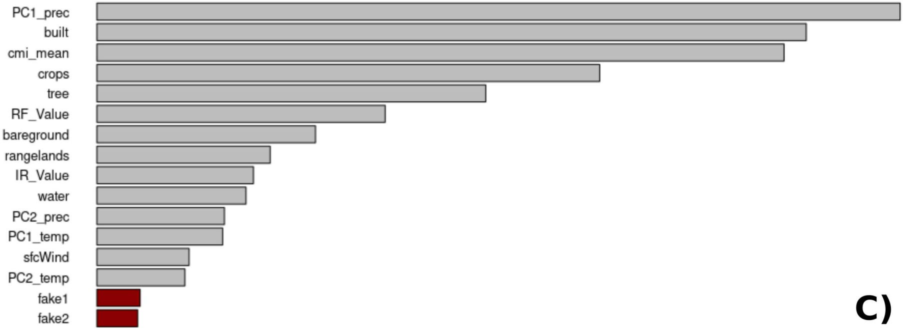
- **Stress resistance** : *CG30015* (heavy metals), *Olf413*, *Prosalpha2* (pesticides)
- **Locomotion / dispersion** : *CG34215*, *Oamb*
- **Chemosensation and feeding behavior** : *Cheb42c*, *Gr57a*

Understand past invasion and predict future ones



Untransformed variables related to **land use** (in general) and **perennial crops abundance**.

Variable importance



- All variables relevant to adaptation (i.e. higher importance than fake variables)
- Highest impact for precipitation and land use.

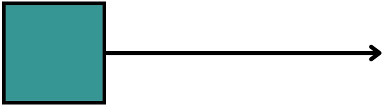
Understand past invasion

$\hookrightarrow$ Did the invaded areas pose an adaptive challenge ?

Source environment

Fraimout et al. (2017)

Liaoyuang, China



Invaded environments

Understand past invasion

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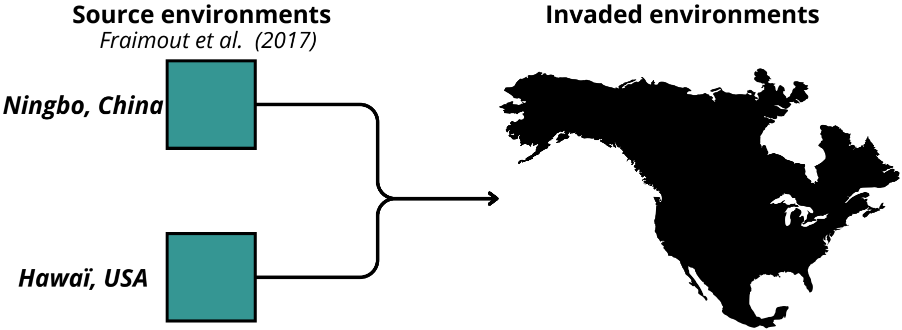
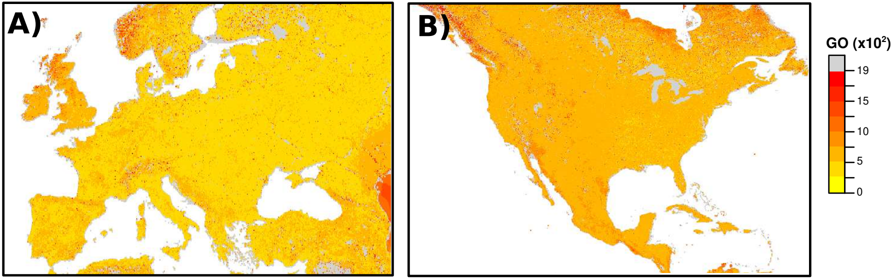


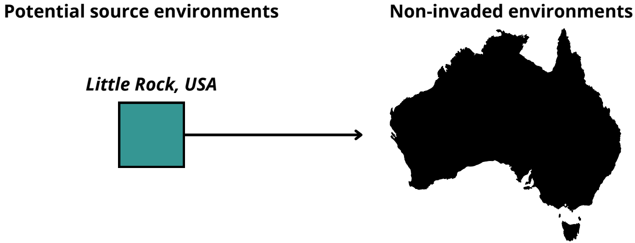
Understand past invasion



⟶ Low to moderate, and homogeneous, adaptive challenge in Europe and (to lower extent) North America

Predict future invasions

$\hookrightarrow$ Which yet-to-be invaded areas are susceptible to invasion ?



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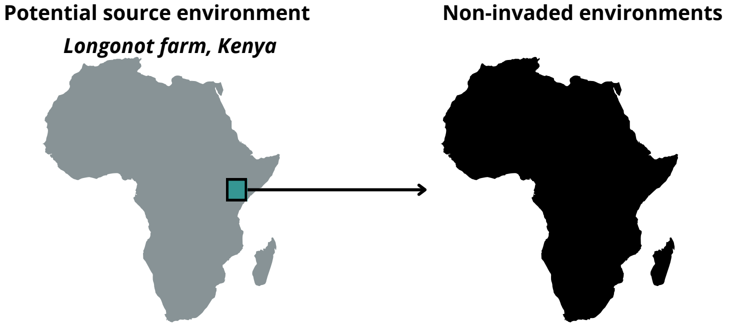

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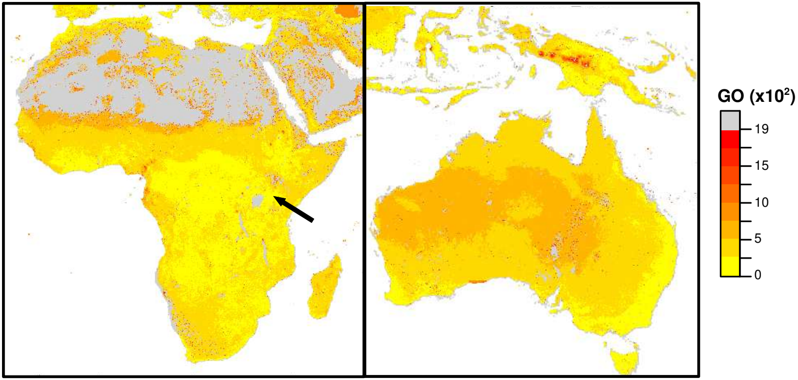
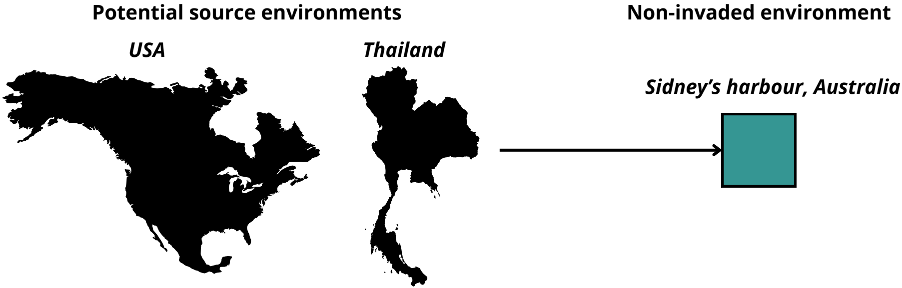


⟶ Large areas with a low risk of maladaptation
⟶ Biologically consistent values

Predict future invasions

$\hookrightarrow$ Which populations are most likely to invade at risk areas ?



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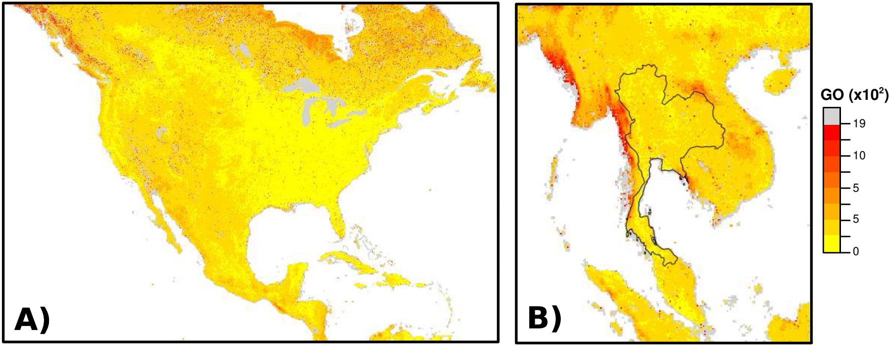
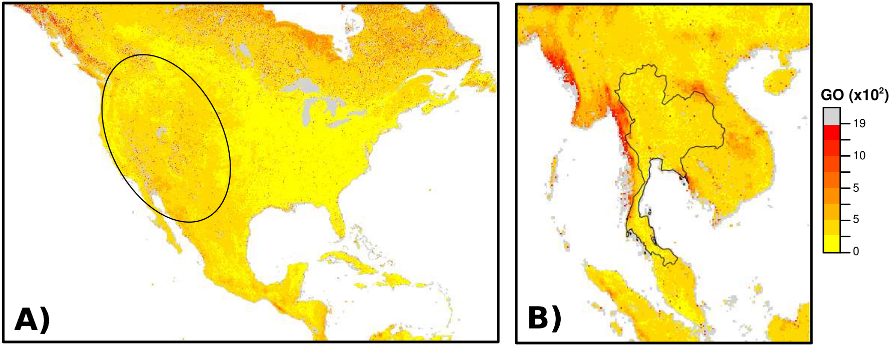


⟶ Low to intermediate values

Predict future invasions



⟶ Low to intermediate values

⟶ Thailand presents a greater risk, but...

⟶ Fruits in the USA are more infested

Conclusions

A better understanding of the past invasion of *D. suzukii*

- Many genes related to e.g. stress resistance, locomotion or chemosensation may have contributed to rapid adaptation in the invaded areas (polygenic signal).
- BUT potential pre-adaptation to invaded environments.

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A tool to anticipate future invasions

- Predictions consistent with the biology of the species
- Large areas still at a low risk of maladaptation

Conclusions

A better understanding of the past invasion of *D. suzukii*

- Many genes related to e.g. stress resistance, locomotion or chemosensation may have contributed to rapid adaptation in the invaded areas (polygenic signal).
- BUT potential pre-adaptation to invaded environments.

A tool to anticipate future invasions

- Predictions consistent with the biology of the species
- Large areas still at a low risk of maladaptation

However...

- Influence of source population choice ? Admixture ? Number of invading individuals ?

Outline

Genomic offset and invasive species

GO is a promising tool to study biological invasions :

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- A wide range of possible uses

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Open questions

- Include invasive populations in training ?
- Effect of admixture ? Inbreeding ?

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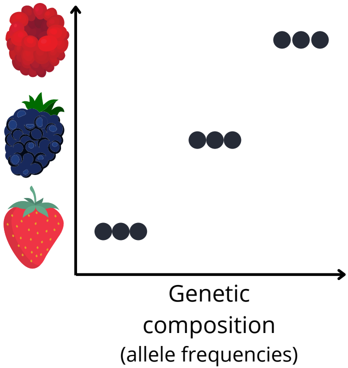
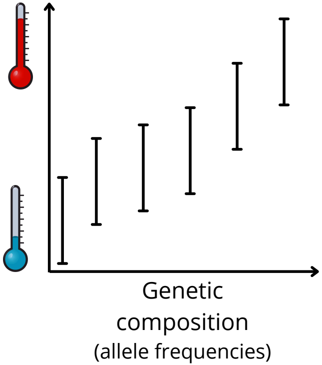
A need to work on

- GEA robustness to correlations among variables / SNPs
- Interpreting the absolute (rather than relative) GO value : definition of thresholds not to be exceeded (Lachmuth et al., 2023) ? Credibility intervals ?

One alternative : predicting the environment ?





Acknowledgements

Thank you for your attention !

- Camus et al (2024), Evolutionary applications, 17(6), e13709.
- Camus et al (2025), BioRxiv